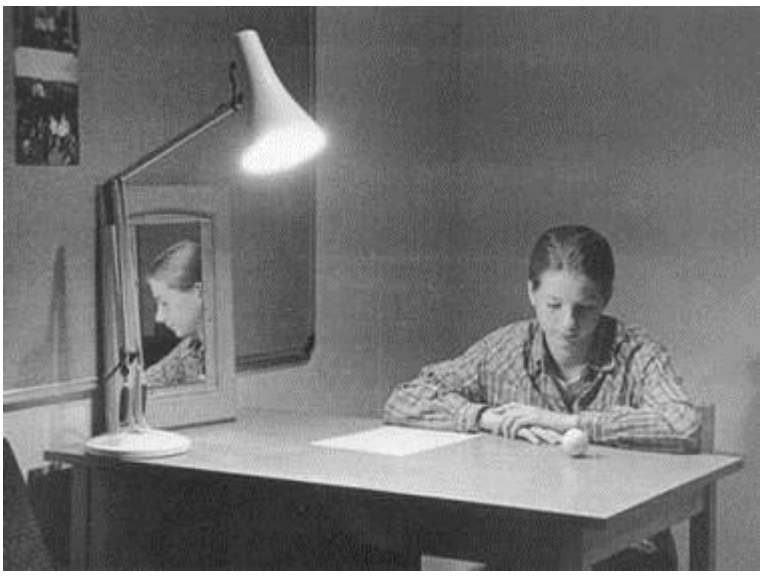
Q1.

Light shines onto a ball. Naomi is looking at the ball.



(a) Describe how light from the lamp lights up the ball and makes it visible to Naomi.

.....

.....

.....

.....

2 marks

(b) (i) Naomi uses different colours of light and different coloured balls.

Complete the table to show the colours that the balls appear to Naomi.

colour of ball	colour of the light	the colour the ball appears to Naomi
white	red	
green	white	

2 marks

(ii) Why does a black object appear black in any light?

.....

1 mark

(c) Choose from the following terms to complete the sentences below.

less than **equal to** **greater than**

At a plane mirror, the angle of incidence is .....

the angle of reflection. The distance from the object to the mirror is

..... the apparent distance from the mirror to the image.

2 marks

(d) A beam of white light shines onto a sheet of white paper. An identical beam of light shines onto a mirror. The light is scattered from the paper and reflected from the mirror.

Describe how scattering by paper and reflection by a mirror are **different** from each other.

.....

.....

.....

.....

2 marks

Maximum 9 marks

Q1.

(a) the light is scattered by the ball

*accept 'it is scattered **or** reflected*
***or** bounces off the ball'*

1 (L5)

some of the light from the ball enters Naomi's eye

*accept 'it goes into **or** gets to her eye'*

1

(b) (i)

colour of ball	**colour of the light**	**the colour the ball appears to Naomi**		
white	*red*	red	*do not accept 'pink' or 'light red'*	**1 (L6)**
green	*white*	green	*do not accept 'light green'*	**1 (L6)**

(ii) any **one** from

- it absorbs all the light
  accept 'it absorbs light'
- it does not scatter any light
  accept 'it does not reflect light'

1 (L6)

(c) equal to

1 (L6)

equal to

*accept 'equals' **or** 'the same as'*

1 (L6)

(d) **one mark is for describing scattering and one mark is for describing reflection**

scattering sends **or** reflects light in all directions

accept 'scattered light goes all over the place'
***or** 'the light from the paper goes off in lots of rays'*
***or** 'no image can be seen in the paper'*

1 (L6)

reflection sends light in one direction **or** to one point

accept 'the light from the mirror is all in one ray or beam'
***or** 'reflected light goes at one exact angle'*
***or** 'an image can be seen in the mirror'*

1 (L6)

[9]